A counterexample to Kruglov necessity for embeddings of generalized Rosenthal spaces

Candidate counterexample for arXiv:2006.00936

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Status. Candidate counterexample, likely valid, requiring human review. In the unrestricted formulation printed in the source, the Kruglov property is not necessary: take $E = L^\infty[0,1]$.

1 Source problem

For a rearrangement-invariant Banach function space E on $[0,1]$, Astashkin and Curbera define a rearrangement-invariant space Z_E on $(0,\infty)$ and then define the generalized Rosenthal space $\mathcal{U}_E$ as the closed linear span in Z_E of a countable sequence of characteristic functions $\{\chi_{A_n}\}_{n=1}^\infty$ satisfying their conditions (2). Their notation $Y \hookrightarrow X$ means that there is an arbitrary isomorphic embedding from Y into X.

The source says on PDF page 4:

> "In the concluding part of the paper, in Theorem 6, we prove a partial result related to the problem if the Kruglov property of a r.i. space E is a necessary condition for the existence of an isomorphic embedding $T : \mathcal{U}_E \to E$."

The stated partial result assumes that images of suitable canonical block vectors are independent symmetrically distributed random variables satisfying an additional uniform lower-tail condition [1, Theorem 6].

> In the concluding part of the paper, in Theorem 6, we prove a partial result related to the problem if the Kruglov property of a r.i. space E is a necessary condition for the existence of an isomorphic embedding $T : \mathcal{U}_E \to E$. We consider the case when T sends the basis functions χ_{A_n}, $n = 1, 2, \ldots,$ of Z_E to some independent symmetrically distributed r.v.'s in E.

Figure 1: The problem and the scope of the source's partial result on PDF page 4 of [1].

2 Proof intuition

The issue is that an arbitrary Banach-space embedding need not retain any of the probabilistic structure of the canonical basis. The space $\mathcal{U}_E$ is always separable, simply because it is generated by a countable sequence. At the opposite extreme, $L^\infty[0,1]$ is a universal host for separable Banach spaces, so it automatically contains a copy of its own $\mathcal{U}_E$.

But L^∞ is too close to the endpoint to have the Kruglov property: the Kruglov transform of the constant one function is an unbounded Poisson random variable. These two elementary observations give the counterexample.

3 Two elementary embedding facts

Lemma 1. *For every rearrangement-invariant space E in the source construction, $\mathcal{U}_E$ is separable.*

Proof. By definition, $\mathcal{U}_E$ is the closed linear span of the countable sequence $\{\chi_{A_n}\}_{n=1}^\infty$. Finite linear combinations using a countable dense subfield of the scalars form a countable dense subset of this closed span. $\square$

Lemma 2. *Every separable Banach space embeds linearly isometrically into $L^\infty[0,1]$.*

Proof. Let X be separable. If $X = \{0\}$ there is nothing to prove. Choose a dense sequence (x_n) in the unit sphere of X. By Hahn–Banach, choose $x_n^* \in X^*$ with

$$\|x_n^*\| = 1, \qquad x_n^*(x_n) = 1.$$

In the complex case the phase may be chosen so that the second equality is literal. Define

$$J : X \longrightarrow \ell_\infty, \qquad Jx = (x_n^*(x))_{n=1}^\infty.$$

This map is linear and contractive. If $\|x\| = 1$ and $x_{n_k} \to x$, then

$$|x_{n_k}^*(x)| \geq |x_{n_k}^*(x_{n_k})| - \|x - x_{n_k}\| = 1 - \|x - x_{n_k}\| \longrightarrow 1.$$

Therefore $\|Jx\|_\infty = 1$, and homogeneity shows that J is an isometry.

Now choose a measurable partition $(B_n)_{n=1}^\infty$ of $[0,1]$ with $m(B_n) > 0$ for every n, and define

$$S : \ell_\infty \longrightarrow L^\infty[0,1], \qquad S((a_n)) = \sum_{n=1}^\infty a_n \chi_{B_n}.$$

Every coordinate value occurs on a set of positive measure, so $\|S((a_n))\|_{L^\infty} = \sup_n |a_n|$. Hence S and $S \circ J$ are linear isometries. $\square$

4 The counterexample

Recall that E has the *Kruglov property* if $f \in E$ implies $\pi(f) \in E$, where

$$\pi(f) = \sum_{i=1}^N f_i,$$

the f_i are independent copies of f, and N is an independent Poisson random variable with parameter one.

Theorem 1. *The Kruglov property is not necessary for the existence of an isomorphic embedding $\mathcal{U}_E \to E$ in the unrestricted formulation of [1]. More precisely, the rearrangement-invariant space*

$$E = L^\infty[0,1]$$

has the Fatou property and satisfies $\mathcal{U}_E \hookrightarrow E$, but it does not have the Kruglov property.

Proof. Lemma 1 says that $\mathcal{U}_{L^\infty}$ is separable. Applying Lemma 2 with $X = \mathcal{U}_{L^\infty}$ gives a linear isometric embedding

$$\mathcal{U}_{L^\infty} \longrightarrow L^\infty[0,1].$$

Thus the required isomorphic embedding exists. The familiar monotone-limit property of the essential-supremum norm shows that L^∞ has the Fatou property.

To disprove the Kruglov property, take $f = \mathbf{1}$. Every independent copy of f is again the constant one function, and hence

$$\pi(\mathbf{1}) = N.$$

For every integer $m \geq 0$,

$$\mathbb{P}(N = m) = \frac{e^{-1}}{m!} > 0.$$

Consequently N is essentially unbounded. Thus $\mathbf{1} \in L^\infty[0,1]$ while $\pi(\mathbf{1}) \notin L^\infty[0,1]$, and L^∞ has no Kruglov property. □

Remark 1 (Scope of the counterexample)**.** *The embedding above is deliberately arbitrary; it need not send canonical vectors to independent random variables or satisfy the lower-tail hypothesis of [1, Theorem 6]. There is therefore no conflict with that partial theorem. If the intended question silently restricts E to separable or order-continuous r.i. spaces, or restricts T to preserve canonical probabilistic structure, that strengthened question remains outside the present result. Those restrictions do not occur in the printed introduction statement.*

5 Verification and bounded novelty check

The proof is noncomputational. Its only ingredients are the source definition of $\mathcal{U}_E$, a countable Hahn–Banach norming family, a positive-measure partition of $[0,1]$, and the elementary Poisson law. The accompanying `verification.md` checks each step and the exact scope.

On 11 August 2026, the local run indexes and downloaded arXiv corpus were searched by arXiv id and the terms "Kruglov property", "$\mathcal{U}_E$", "isomorphic embedding", and "necessary". External exact and close searches used the same phrases, the paper title and authors, and the journal DOI. They found the source and earlier sufficient-condition literature but no explicit answer or this L^∞ counterexample. OpenAlex listed no citing works for the DOI at the time of checking; MaRDI listed five related citing titles, none of which surfaced an answer in the bounded search. Novelty confidence is moderate and requires specialist confirmation.

Human-review recommendation. Confirm the unrestricted meaning of $\hookrightarrow$ and decide whether the authors intended an unstated separability or structure-preserving condition. No mathematical gap is known for the literal formulation.

References

[1] S. V. Astashkin and G. P. Curbera, *Rosenthal's space revisited*, Studia Math. **262** (2022), no. 2, 197–224, doi:10.4064/sm201011-4-1; arXiv:2006.00936.